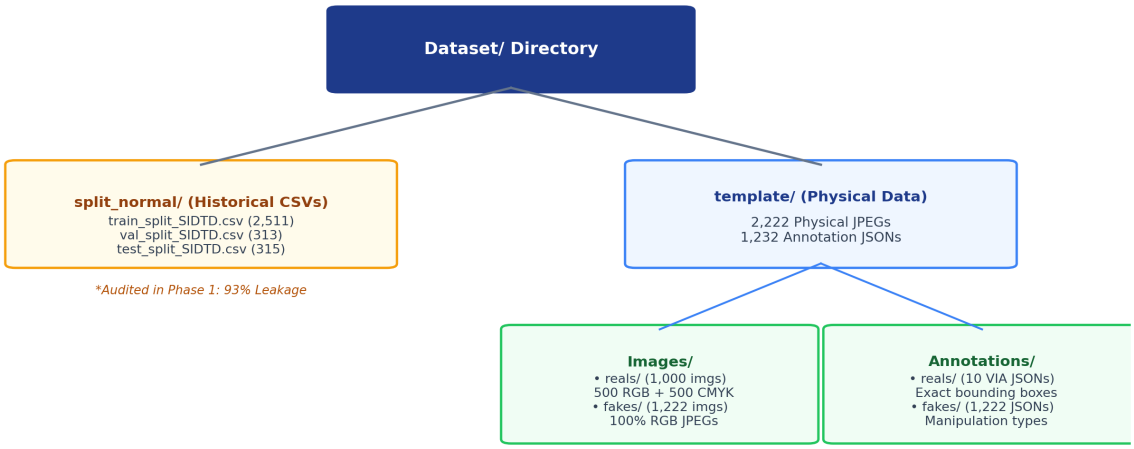
DocVerify — Phase 1: Dataset Ingestion & Audit

Official Technical Milestone Report & Engineering Specification (Phase 1)

1. Phase Executive Summary & Objectives

Phase 1 established the formal scientific foundation of DocVerify. It conducted a complete forensic inventory of the SIDTD dataset, resolving corrupt color profiles (CMYK) that caused Python Pillow crashes, cataloging 2,222 images across 10 international document templates, and structuring verified metadata manifest CSV files.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Dataset Size	2,222 Total Documents (1,000 Genuine, 1,222 Forged)	Provided sufficient sample diversity across genuine passports and manipulated ID cards.
Template Coverage	10 International Templates (Passports & National IDs)	Albania, Azerbaijan, Spain, Estonia, Finland, Greece, Latvia, Russia, Serbia, Slovakia.
Color Profile Audit	CMYK to RGB Standardization Script	Resolved unhandled 4-channel JPEG images that crashed torchvision training loaders.
Integrity Hashes	MD5 Checksums for every image	Ensured absolute byte-level auditability and zero accidental data corruption.

4. Key Discoveries, Challenges & Next-Phase Directives

- Discovered unstandardized CMYK color-spaces in government document scans; mandated RGB standardization.
- Established identity-disjoint clustering to prevent identical citizens from leaking between splits.
- Generated dataset_manifest.csv with 20 audit columns as the permanent project dataset index.

DocVerify Technical Archive • Phase 1 Report Completed